

2(a)(i)	50 (cm ³) ;	1
2(a)(ii)	7 (straws) ;	1
2(b)	<i>any 4 of:</i> <i>estimate of numbers:</i> 1. calculate the volume of the sample or volume of sample is, (for mm³) $0.25 \times 0.25 \times 0.1 / 0.00625 / 6.25 \times 10^{-3}$ / (for cm³) $0.025 \times 0.025 \times 0.01 / 0.00000625 / 6.25 \times 10^{-6}$; 2. divide the number of sperm by the volume and method of converting to cm³ ; <i>deciding which sperm to include:</i> 3. <i>idea of:</i> count those where sperm (fully) inside the square / 0.0625 (mm²) / all (small) squares / haemocytometer / quadrant / grid ; 4. <i>idea of:</i> include those touching the (line) top and left / up and right or exclude those touching bottom and right / down and left ; <i>detail of method:</i> 5. AVP ; e.g. <i>ref to</i> dilution of sample / stirring / even spreading / immobilising sperm / use of coverslip / screenshot for counting ;	4
2(c)	<i>sugar: idea that</i> (it) provides or gives energy / is an energy source / (is used) in respiration / (is used) to produce ATP ; <i>buffer:</i> (change in pH may), denature enzymes / proteins ; <i>antibiotics: idea of,</i> killing / stopping reproduction of, bacteria / AW ;	3